Last time:

- $f_n$ integrable
- $f_n$ converges pointwise to $f$

$\Big\} \not\Rightarrow \int_a^b f_n \to \int_a^b f$

**Thm** Let $f_n, f : [a,b] \to \mathbb{R}$. If $f_n$ is integrable $\forall n \in \mathbb{N}$ and $f_n$ converges uniformly to $f$, then:

① $f$ is integrable.

② $\lim_{n\to\infty} \int_a^b f_n = \int_a^b f$.

Pf of ②: Fix $\varepsilon > 0$. Want: $\exists N$ s.t.

$$\left| \int_a^b f_n - \int_a^b f \right| < \varepsilon \qquad \forall n \geq N.$$

As $f_n$ converges uniformly to $f$, $\exists N$ s.t.

$$|f_n(x) - f(x)| < \frac{\varepsilon}{2(b-a)} \qquad \forall x \in [a,b], \quad \forall n \geq N.$$

So, for $n \geq N$ we have

$$f_n(x) - f(x) < \frac{\varepsilon}{2(b-a)} \quad \Rightarrow \quad f_n(x) < f(x) + \frac{\varepsilon}{2(b-a)}$$

$$\Rightarrow \int_a^b f_n(x)\,dx \leq \int_a^b f(x) + \frac{\varepsilon}{2(b-a)}\,dx$$

$$= \int_a^b f(x)\,dx + \frac{\varepsilon}{2(b-a)} \int_a^b 1\,dx$$

$$= \int_a^b f(x)\,dx + \frac{\varepsilon}{2\cancel{(b-a)}} \cdot \cancel{(b-a)}$$

Similarly,

$$f_n(x) - f(x) > -\frac{\varepsilon}{2(b-a)} \quad \Rightarrow \quad \ldots$$

$$\Rightarrow \int_a^b f_n(x)\,dx \geq \int_a^b f(x)\,dx - \frac{\varepsilon}{2}$$

Together:

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \frac{\varepsilon}{2} < \varepsilon \qquad \forall n \geq N.$$

So $\int_a^b f_n \to \int_c^b f$ as $n \to \infty$. $\square$

- When does $\int \sum f_n = \sum \int f_n$ ?

<u>Cor</u> Let $f_n, f : [a,b] \to \mathbb{R}$. If $f_n$ is integrable $\forall n \in \mathbb{N}$ and the series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly, then $f(x) = \sum_{n=1}^{\infty} f_n(x)$ is integrable and

$$\int_a^b \sum_{n=1}^{\infty} f_n \, dx = \sum_{n=1}^{\infty} \int_a^b f_n \, dx.$$

<u>Pf</u>: Apply the previous theorem to $s_n(x) = \sum_{k=1}^{n} f_k(x)$. $\square$

- Example: power series
- Can we integrate a power series term-by-term?

<u>Thm</u> If the power series $f(x) = \sum_{n=0}^{\infty} c_n x^n$ has radius of convergence $R > 0$, then

$$\int_0^x f(t)\, dt = \sum_{n=0}^{\infty} \frac{c_n}{n+1} x^{n+1} \qquad \forall x \in (-R, R).$$

<u>Pf</u>: Fix $x \in (-R, R)$.

Case: $x < 0$. Recall that the series $\sum_{n=0}^{\infty} c_n t^n$ converges uniformly to $f(t)$ on $[x, 0]$ (by Lecture 35). So, by the corollary,

$$\int_x^0 f(t)\,dt = \sum_{n=0}^{\infty} \underbrace{\int_x^0 c_n t^n\,dt}$$

$$= c_n\left(\frac{0^{n+1}}{n+1} - \frac{x^{n+1}}{n+1}\right)$$

$$= -\frac{c_n}{n+1}x^{n+1}$$

$$\Rightarrow \int_0^x f(t)\,dt = -\int_x^0 f(t)\,dt = \sum_{n=0}^{\infty} \frac{c_n}{n+1}x^{n+1}.$$

Case: $x>0$. Similar, but for $[0,x]$. $\square$

- Can we differentiate a power series term-by-term?

<u>Cor</u> If the power series $f(x)=\sum_{n=0}^{\infty} c_n x^n$ has radius of convergence $R>0$, then $f:(-R,R)\to\mathbb{R}$ is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} nc_n x^{n-1} \quad \forall x\in(-R,R).$$

<u>Pf</u>: Claim: The radius of convergence of $\sum nc_n x^{n-1}$ is also $R$. Note that $\sum nc_n x^{n-1}$ and $\sum nc_n x^n$ have the same radius of convergence; $\sum nc_n x^n$ is $x$ times $\sum nc_n x^{n-1}$, so they converge for exactly the same values of $x$. Now, for $\sum nc_n x^n$,

$$\limsup |nc_n|^{\frac{1}{n}} = \limsup (\underbrace{n^{\frac{1}{n}}}_{\to 1}\cdot|c_n|^{\frac{1}{n}}) = 1\cdot\limsup |c_n|^{\frac{1}{n}}$$

$$\Rightarrow \frac{1}{\limsup |nc_n|^{\frac{1}{n}}} = \frac{1}{\limsup |c_n|^{\frac{1}{n}}} = R$$

By the claim, we can apply the previous theorem to $g(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ on $(-R, R)$:

$$\int_0^x g(t)\, dt = \underset{\displaystyle f(x) - c_0}{\overset{\displaystyle \sum_{n=1}^{\infty} c_n x^n}{\parallel}} \quad \forall x \in (-R, R)$$

$$\Rightarrow \quad f(x) = c_0 + \int_0^x g(t)\, dt$$

$$\overset{FTC}{\Rightarrow} \quad f'(x) = 0 + g(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}. \qquad \square$$

<u>Ex</u> Recall the geometric series

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \qquad \text{for } x \in (-1, 1).$$

① Integrating, we get (by the previous theorem):

$$\sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1} = \int_0^x \frac{1}{1-t}\, dt = -\ln(1-x)$$

$$\Rightarrow \quad \ln(1-x) = -\sum_{n=1}^{\infty} \frac{1}{n} x^n$$

$$\Rightarrow \quad \ln(1+x) = -\sum_{n=1}^{\infty} \frac{1}{n} (-x)^n$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n \qquad \text{for } x \in (-1, 1).$$

② Differentiating, we get (by the corollary):

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2} \qquad \text{for } x \in (-1, 1)$$

- This allows us to compute lots of new power series from old ones.

<u>In summary</u>:

1. $\{f_n\}$ converges uniformly on $[a,b]$ $\Rightarrow$ $\lim\limits_{n\to\infty} \int_a^b f_n = \int_a^b \lim\limits_{n\to\infty} f_n$

2. $\sum f_n$ converges uniformly on $[a,b]$ $\Rightarrow$ $\sum\limits_{n=1}^{\infty} \int_a^b f_n = \int_a^b \sum\limits_{n=1}^{\infty} f_n$

3. This applies to $\sum c_n x^n$ when $|x| <$ radius of convergence.

<u>Rmk</u> The direct analogue of 1 for derivatives is false. E.g.:

$f_n(x) = \frac{1}{n}\sin(nx)$ converges uniformly to $f(x) = 0$,

$f_n'(x) = \cos(nx)$ does not converge uniformly to $f'(x) = 0$.